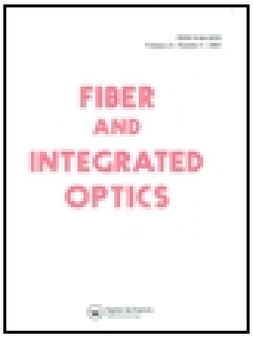

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Performance Analysis on Single- and Double-Pass Ytterbium-Doped Fiber Amplifier in the 1 µm Region

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ABSTRACT

In this work, single- and double-pass configurations of ytterbium-doped optical fiber amplifiers (YDFAs) have been investigated. This low quantum defect rare-earth dopant and their performance near 1 µm region has been comprehensively investigated on the basis of gain and noise figure in relation with input signal power and input pump power. The obtained results showed that at an input pump power of 100 mW, the gain was proportionally increased with ytterbium-doped fiber (YDF) length for both proposed configurations, but double-pass regime scored higher gain. This is due to the double-propagation of the forwarded and amplified spontaneous emission (ASE) signal into the active medium and hence will maximize the attainable gain near the 1 µm wavelength region. At YDF length of 5 m, the double-pass technique showed better amplification performance at a lower input signal power of −20 dB compared to the higher input signal power of 0 dB. The maximum gain and optical signal-to-noise ratio (OSNR) attained by double-pass configuration were 24.6 dB and 53.4 dB, respectively. Double-pass amplification at input signal power of −20 dB has increased the gain by 11.2 dB in comparison with the single-pass technique.

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KEYWORDS
Fiber amplifier; YDFA; double-pass amplifier

Introduction

Among the unique and outstanding properties of fiber lasers are the excellent beam quality and reduced heat load [1–5]. Furthermore, high power fiber lasers had found potential applications in many aspects of science and technology such as material processing, coherent beam combination, and nonlinear frequency conversion [6–8]. Basically, the two approaches to produce high output average power are either scaling up the power in fiber laser oscillators [9] or adopting the scheme of master oscillator–power amplifiers (MOPA) [10]. The diffracted limited beam quality produced from fiber lasers can produce higher intensity but, on the other hand, it may cause damage to the fiber facet. Mitigation of the many vital limitations imposed on power scaling in fiber laser oscillators such as nonlinear effects is a challenge [11].

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Different strategies and approaches have been proposed and implemented such as tandem pumping [12] getting benefiting from the reduced quantum defect and the possibility to pump close to the emission wavelength in Yb fiber. The other approach using optical amplifiers which can be achieved by single, double, and multi-pass amplification to scale up the power of the seed laser is more practical to obtain sufficient average output power [13, 14]. Though multi-pass amplifiers have imperative practical significance to maximize the magnification of optical fiber amplifiers, it increase the cost and complexity of the system [15]. Double-pass fiber amplifiers might play a great role to overcome these problems. It can effectively reduce the amplified spontaneous emission (ASE), minimize the noise, and increase the gain to achieve high power output [16, 17]. In recent, many kinds of optical fiber amplifiers are developed such as semiconductor optical amplifiers (SOAs) [18] and Raman amplifiers (RAs) [19]. Though they have relatively high gain, SOAs suffer from a high noise figure and high cross-talk level due to four-wave mixing nonlinearity [20]. RAs are frequently used for optical amplification too in near-infrared region due to their high gain and wide bandwidth range. Unfortunately, RAs demand high input pump power and sophisticated gain control [21]. Erbium doped-fiber amplifiers (EDFAs) represent the best alternative that found important applications in wavelength-division multiplexing (WDM) for communications systems at 1530 nm [22, 23].

Lately, ytterbium-doped fiber amplifiers (YDFAs) had attracted vast research attention due to their low quantum defect, high output power, broad-gain bandwidth, and high electrical to optical power conversion [24, 25]. These features make them suitable for medium and high power applications such as free-space optical communication [26], fiber sensing [27], ultra-short pulse amplification [28], and mass manufacturing [29]. Basically, YDFA works on stimulated emission principle, and thus pumping laser is required to excite ytterbium ions from lower to an upper energy level and consequently emits photon energy at the same input signal wavelength [30].

In this work, single- and double-pass YDFAs are designed and simulated at a wavelength of 1.1 μm. We focused on the effect of ytterbium-doped fiber (YDF) length and input pumping power on the most crucial amplifier parameters: gain, noise figure (NF), and optical signal-to-noise ratio (OSNR).

Theory

Technically, the numerical modeling of YDFA is not complicated as long as the ASE is assumed to be homogeneously broadened. The rate equations governing the ytterbium upper and lower state population n_2 and n_1, respectively, can be described as follows [24]:

$$\frac{dn_2}{dt} = n_1(R_{12} + W_{12}) - n_2(R_{21} + W_{21} + A_{21}) \tag{1}$$

$$\frac{dn_1}{dt} = -n_1(R_{12} + W_{12}) + n_2(R_{21} + W_{21} + A_{21}) \tag{2}$$

So, the steady-state values are:

$$n_2 = \frac{R_{12} + W_{12}}{R_{12} + R_{21} + W_{12} + W_{21} + A_{21}} \tag{3}$$

$$n_1 = 1 - n_2 \tag{4}$$

And the transition rates:

$$R_{12} = {}^{p}_{12}I_P/_{h\nu_P} \tag{5}$$

$$R_{21} = {}^{p}_{21}I_P/_{h\nu_P} \tag{6}$$

$$W_{12} = \sigma^s_{12}I_s/_{h\nu_s} \tag{7}$$

$$W_{21} = \sigma^s_{21}I_s/_{h\nu_s} \tag{8}$$

where I_p and I_s are the pump and signal intensities, σ^p_{12} and σ^p_{21} are the effective absorption and emission cross-sections at input pump power wavelength, σ^s_{12} and σ^s_{12} are the effective absorption and emission cross-sections at the signal wavelength, respectively.

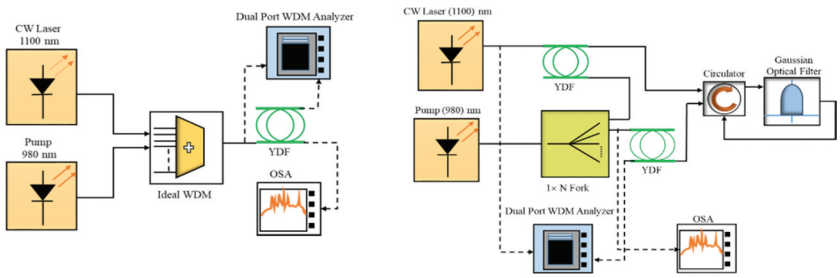
The gain and loss at both pump and signal at a certain position z along single-mode fiber can be given as follows, assuming that N_{tot} is the total density of ytterbium ions:

$$\frac{dP_P}{dz} = \eta_P\left(\sigma^p_{21}n_2 - \sigma^p_{12}n_1\right)N_{tot}P_P \tag{9}$$

$$\frac{dP_s}{dz} = \eta_s\left(\sigma^s_{21}n_2 - \sigma^s_{12}n_1\right)N_{tot}P_s \tag{10}$$

Simulation setup

The proposed models are simulated by Optisys 7 as presented in Figure 1. The YDFA setup comprises YDF with doping concentration of $1{*}10^{+25}$ m^{-3}, numerical aperture of 0.2, and doped-core radius of 1.7 μm. The gain medium is pumped by 980 nm laser source with maximum output power of 100 mW. The input signal is seed by an 1100 nm continues-wave (CW) laser. Optical



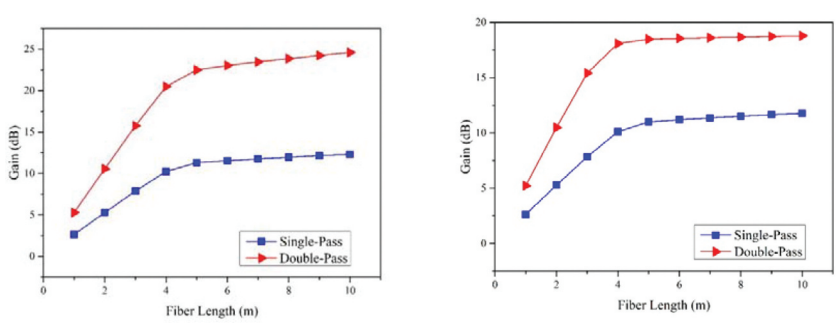
(a) (b)

Figure 1. Simulated YDFA at 1100 nm: (a) single-pass regime, (b) double-pass regime.

WDM of 0 dB loss is used to combine 980 and 1100 nm wavelengths. To insure one direction propagation, an optical circulator is incorporated. To optimize the output wavelength, a Gaussian optical filter with 0.8 nm bandwidth is added to the cavity. Dual-port WDM optical analyzer is connected to the setup to measure both gain and noise figure at the same time. Optical spectrum analyzer (OSA) of 0.01 nm resolution is used to examine the spectrum of the output signal. The single-pass is modified to double-pass configuration where two YDFs of the same mentioned specifications are incorporated as shown in Figure 1b.

Results and discussions

The amplification gain as a function of YDF length for the single and double-pass configurations at input signal power of −20 dB and 0 dB is shown in Figure 2. Obviously, for single-pass and double-pass configurations, the amplification was increasing rapidly with the YDF length to approximately 5 m. For longer fiber lengths, the amplification of the gain medium gets saturated. At



(a) (b)

Figure 2. Gain of single- and double-pass YDFAs versus YDF length (a) at input signal power of −20 dB, (b) at input signal power of 0 dB.

YDF length of 5 m, which is the optimum length for both proposed configurations, double-pass configuration shows amplification gain higher than single-pass technique by 11.2 dB at input signal power of −20 dB and 7.5 dB at input signal power of 0 dB. Figure 3 illustrates the noise figure (NF) of both proposed configurations at input signal power of −20 dB and 0 dB.

The NF in single-pass amplification is recognized to be higher compared to the double-pass configuration for low and high input signal powers. This is because of unproductive population inversion that happened when there are insufficient atoms being pumped to the higher excitation levels. Subsequently, this will lead to weak stimulated emission and thus NF increment. Figure 4 shows the gain and NF versus the input pumping power results of the simulated single- and double-pass YDFA configurations at the optimum fiber length. By using the 1100 nm signal and sweeping the input pumping power from 0 mW to 100 mW for input signal power of 0 dB, the maximum gain for single and double-pass amplification are 10.99 and 18.48 dB at 100 mW, respectively.

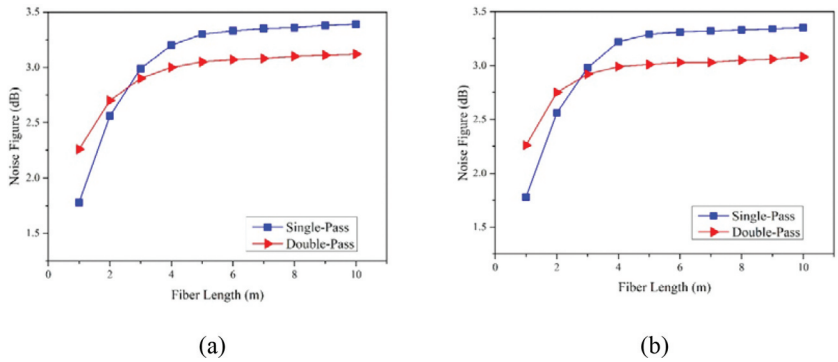


Figure 3. NF of single- and double-pass YDFAs versus YDF length (a) at input signal power of −20 dB, (b) at input signal power of 0 dB.

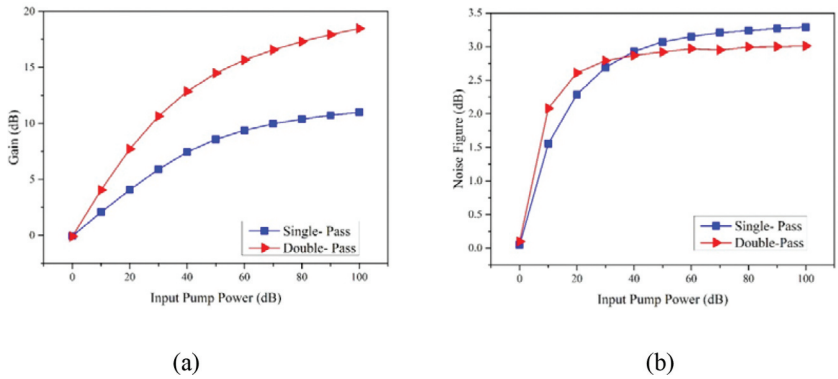


Figure 4. Performance of single- and double-pass YDFAs at input signal of 0 dB: (a) gain versus input pump power, (b) noise figure versus input pump power.

Furthermore, the gain enhancement of 7.49 dB is observed in the double-pass YDFA. The gain enhancement is the difference between the recorded maximum gain for single- and double-pass configurations. This gain enhancement is a result of double propagation of the signal in the YDF which increases the effective length of the amplifier and thus the gain.

The output power as a function of input pumping power for the simulated regimes at the optimum YDF length and signal power of 0 dB is shown in Figure 5. The simulated single- and double-pass YDFAs showed output power growth of 12.4 dB and 17.78 dB, respectively, which are higher than the values recorded by the previously published work by P. Zhang et al. [31]. Figure 6 illustrates the spectra of the proposed configurations at input signal of 0 dB. It is recorded that double-pass YDFA presents an OSNR enhancement of 11.8 dB over that of single-pass YDFA. Furthermore, OSNRs' enhancement 0 f 13.6 and 18.3 dB has been obtained for both single- and double-pass regimes, respectively, compared to the results of reference 31.

Conclusions

The single- and double-pass YDF amplifiers have been designed and simulated at 1100 nm for the first time. Their gain and noise figure at −20 and 0 dB input signal power have been compared, and their performance as a function of

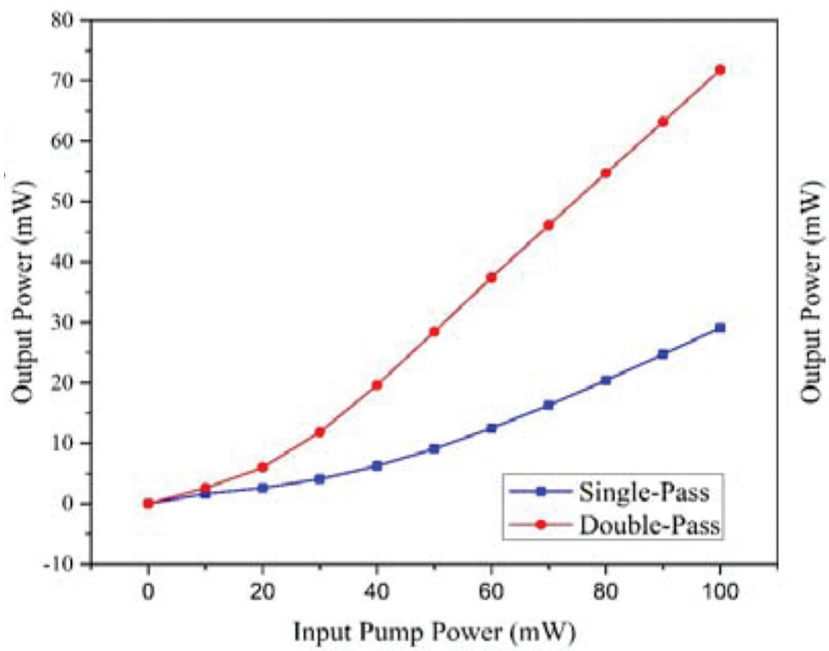
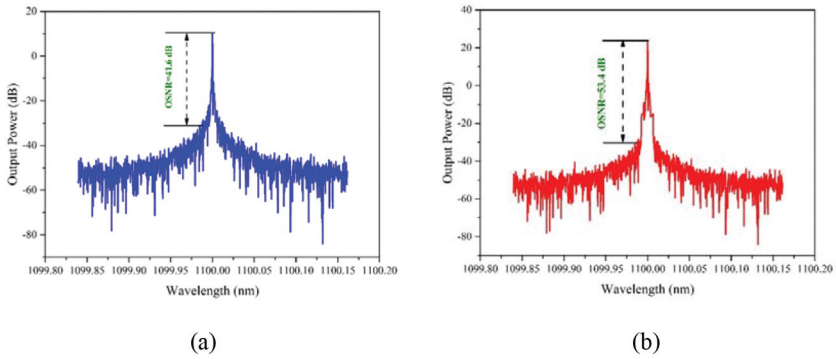


Figure 5. Output power of the proposed YDFAs versus the input pumping power at signal power of 0 dB.



(a) (b)

Figure 6. Spectra of the proposed amplifiers at input signal of 0 dB and input pumping power of 100 mW: (a) single-pass YDFA, (b) double-pass YDFA.

input pump power was investigated. From the simulated results, double-pass amplification showed higher gain, lower NF, and higher OSNR compared with single-pass regime. The maximum gain and OSNR recorded by double-pass configuration were 24.6 dB and 53.3 dB, respectively. At the optimum YDF length, double-pass amplification at input signal power of −20 dB has increased the gain by 11.2 dB in comparison with the single-pass technique, while at higher signal power of 0 dB, the obtained double-pass gain enhancement over single-pass technique was 7.5 dB.

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